THAKUR COLLEGE OF
ENGINEERING & TECHNOLOGY
(Approved by AICTE, Govt. of Maharashtra & Affiliated to University of Mumbai)
(Accredited Programmes by National Board of Accreditation, New Delhi)
ISO 9001 : 2015 Certified • Accredited with A+ Grade by NAAC

IN-SEMESTER EXAMINATION-I

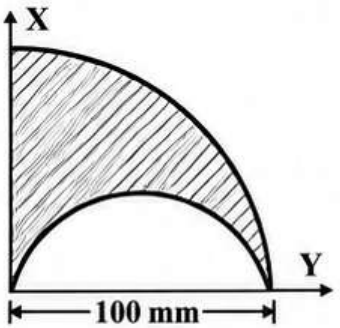
F.E./F.T. (Semester-I)

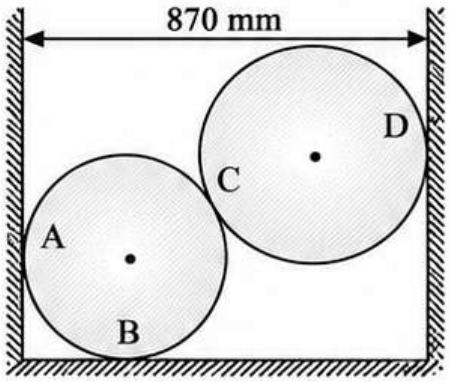
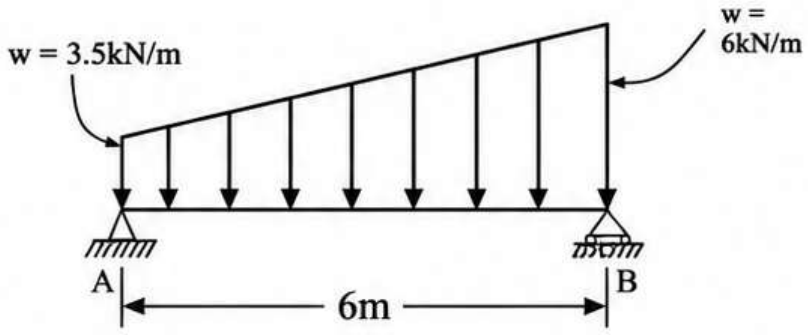
SUBJECT – Engineering Mechanics

Branch: E&TC, E&CS, IT, MECH, M&ME, AI&ML **Date: 29/10/2025**
Div.: ALL **Timing: 12:00 PM to 01:00 PM**
Duration: 60 Minutes **Maximum Marks: 20**

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
	a.	Determine the centroid for a semicircular disc.	2	CO1	R
	b.	Find Magnitude of force F = 8i - 6j - 10k acting on a javelin rod.	2	CO1	A
	c.	Show the reactions acting on the hook fixed in the wall	2	CO2	R
	d.	Draw free body diagram of a football placed in a cylindrical container having same diameter?	2	CO2	R
	e.	Define Poisson's Ratio	2	CO2	R
	f.	Define limiting friction	2	CO3	R
	g.	Define Angle of Friction with neat sketch.	2	CO3	R
Q.2	a.	A force 'P' = 10 N in magnitude acts along direction AB whose coordinates of points A and B are (5, 5, -3) m and (9, 6, 2) m respectively. Another force 'Q' = 5 N in magnitude acts along BC where C has coordinates (-2, 11, -5) m. Determine the resultant of forces.	5	CO1	A
		OR			
	b.	Find the Centroid of the shaded area of the following figure. 	5	CO1	A

Q.3	a.	Two smooth spheres of weight 100 N and radius 250 mm each are in equilibrium in a horizontal channel of width 870 mm as shown in Fig. Find the reactions at the surfaces of contact C & D, assuming all the surfaces to be smooth.  	5	CO2	A
		OR			
	b.	Draw FBD and Find support reaction of the beam as shown in fig.  	5	CO2	An